

Week 2 Task 2: Cybersecurity Documentation

Task 1: SQL Injection Practice

Objective:

To test SQL Injection attacks using Damn Vulnerable Web App (DVWA) and analyze how SQL queries can be manipulated.

Steps Performed:

1. Logged into DVWA with admin credentials.
2. Navigated to the SQL Injection section.
3. Entered a test payload: `1' OR '1'='1` in the input field.
4. The database returned multiple entries, revealing a vulnerability.
5. Verified that unauthorized data retrieval was possible.

Results:

The SQL Injection attack successfully bypassed authentication and retrieved user details from the database.

Task 2: Web Traffic Analysis with Burp Suite

Objective:

To intercept HTTP requests and analyze captured data using Burp Suite Community Edition.

Steps Performed:

1. Installed and configured Burp Suite as a proxy.
2. Enabled interception and captured an HTTP request.
3. Observed request headers, including user-agent and authentication data.
4. Modified request parameters to analyze security loopholes.

Results:

Captured sensitive data, such as credentials and request payloads, which can be exploited in real-world attacks.

Conclusion:

These exercises demonstrated how attackers exploit SQL injection vulnerabilities and intercept sensitive data using proxy tools. Proper security measures, such as input validation and encryption, are necessary to prevent these attacks.

Submission Details:

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